STABLEMAX-PPO: FINITE-BUDGET RLHF WITH A MAX–STABILITY TRADE-OFF

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ABSTRACT

Best-of-N sampling is common when a language model has a finite response budget and the best candidate is selected. We study the resulting terminal- reward RLHF problem with a frozen ordinal Moment RM trained from HelpSteer2 repeated helpfulness ratings and preference pairs. Rather than replacing this bounded distribution by a scalar or Gaussian reward, we derive an exact finite-support Best-of-N objective and a one-sided calibration correction for its upper tail. StableMax-PPO turns the resulting exact marginal credit into a PPO update with mean and independent-quality floors, full-categorical KL measurement, and batch-level rollback. The design exposes a tunable operating point: the method retains competitive finite-budget maximum performance while reducing policy drift relative to mean- and group-relative baselines. Its contribution is therefore a principled max–stability trade-off, not a claim of universal dominance or direct human-preference improvement.

1 INTRODUCTION

Language models are increasingly deployed with a finite generation budget: several responses are sampled and a downstream system returns, reranks, or inspects the best candidate. Optimizing the expected score of one response therefore targets a different functional from the one used at inference. A mean-improving update can leave the chance of a very high-scoring candidate unchanged, while a direct maximum objective can concentrate probability in a small and brittle region of the response distribution.

The second difficulty is evaluator error. A maximum amplifies mistakes in the upper tail of a reward model, exactly where a policy trained against that model has the strongest incentive to exploit them. Existing Best-of-N, max@k, and inference-aware policy-gradient objectives address the finite-budget target but generally use a scalar score or a nominal distribution. They do not, within the same update, expose the finite ordinal thresholds, quantify calibration uncertainty, and protect ordinary quality and policy drift.

Our starting point is a different evaluator interface. The frozen Moment RM is trained from the repeated helpfulness ratings and preference pairs in HelpSteer2 (Wang et al., 2024b) and returns probabilities over the five ratings $\{0, 1, 2, 3, 4\}$. This choice is consequential: its predictive mean is a model-implied ordinary score, while its conditional variance records a model-based signal of disagreement in the repeated-annotation protocol. The full categorical output lets us compute the deployment maximum exactly; the variance is not added as a reward and is not treated as a human judgment of new responses.

We introduce StableMax-PPO, which combines two new pieces. First, a one-sided calibration envelope turns the ordinal distribution into the exact worst case for a finite-budget maximum and yields a closed-form leave-one-out marginal credit. Second, this credit is placed in a constrained PPO controller with an independent quality floor, a reference-policy KL penalty, and exact rollback of the policy and optimizer state when a tentative batch exceeds the hard KL level. The first piece changes what is optimized; the second makes the desired max–stability operating point explicit and measurable.

The novelty is therefore an interface between an evaluator and a policy optimizer, rather than another PPO variant in isolation. The evaluator exposes the five ordinal thresholds that the deployment maximum actually uses; the calibration envelope says how those thresholds should be stressed when the evaluator is uncertain; and the leave-one-out credit turns the resulting group functional into response-level learning signals. The controller then keeps the target, ordinary quality, and policy motion as separately auditable quantities. A single scalar reward weight cannot reveal whether an apparent maximum gain came from better responses, a changed evaluator regime, or simply a larger policy departure.

The distinction between these pieces is important for interpreting the claim. The ordinal envelope is not another scalar reward shaping term: it changes the functional whose gradient is estimated. The controller does not make the maximum objective easier to win; it makes the cost of obtaining a high maximum observable in the same run. In particular, a method that raises the maximum only by moving far from the reference, lowering ordinary quality, or exploiting one evaluator should appear as a different point on the measured frontier. This lets us ask a falsifiable engineering question: can finite-budget maximum performance be improved without paying the full policy-drift cost of direct maximization?

Our contributions are: (1) an exact finite-N robust objective for a bounded ordinal evaluator; (2) an exact Rao–Blackwellized marginal policy credit and its score-function identity; (3) a compact PPO realization with quality/KL safeguards and batch-level rollback; and (4) a frozen, multi-seed evaluation that reports favorable and failed gates. The experiments support a stability-favoring operating point, not universal superiority over max-oriented baselines.

2 Problem Setup

We consider response-level RLHF with a fixed Best-of-N budget. The policy parameter θ is trainable; the ordinal Moment RM ω, scalar Quality RM η, and reference policy π_{ref} are frozen. For $x \sim \mathcal{D}$, responses factor as $\pi_\theta(Y \mid x) = \prod_h \pi_\theta(a_h \mid x, a_{<h})$, and a rollout contains $Y_{1:N} \sim \pi_\theta(\cdot \mid x)$ independently. Each response receives one terminal evaluation. We assume (A1) conditional independence of rating outcomes across candidates, (A2) frozen evaluators, and (A3) the standard differentiability and likelihood-ratio regularity; these are the only assumptions needed below.

Methodology and motivation. The deployment protocol determines the learning target: the policy is not asked to make every sampled response good, but to make a fixed-budget group contain a response that clears high ordinal thresholds. This makes the evaluator–policy interface part of the problem. A scalar reward can be optimized with standard PPO, but it cannot tell whether a group improvement comes from complementary upper-tail mass, a change in ordinary quality, or a larger departure from the reference policy. We therefore keep these quantities separate throughout the objective, update, and evaluation rather than introducing one composite score.

This setup isolates the source of the engineering difficulty. The policy is trained from response-level likelihood ratios, but the deployment quantity is a group statistic over (N) responses and a latent ordinal rating. A useful training signal must therefore preserve both the group interaction and the finite rating thresholds. We keep the evaluators frozen so that changes in the measured frontier can be attributed to the policy update and its safeguards, rather than to a reward model moving during training.

2.1 Frozen ordinal evaluator

The Moment RM predicts $p_{\omega,r}(x,y) = \Pr_\omega(R = r \mid x, y)$ for $r \in \mathcal{R} = \{0,1,2,3,4\}$, with $\sum_r p_{\omega,r} = 1$. We use $\mu_\omega = \sum_r r p_{\omega,r}$ and $\sigma_\omega^2 = \sum_r (r-\mu_\omega)^2 p_{\omega,r}$. Retaining the full distribution, rather than only μ_ω, is essential because a finite-budget maximum depends on every threshold. The HelpSteer2 repeated annotations make σ_ω^2 a model-based disagreement signal under that protocol, not epistemic uncertainty or a stand-alone bonus. The complete ordinal-probit parameterization is in Appendix A.

This evaluator choice is part of the contribution: it supplies a finite, auditable rating space in which the deployment maximum can be computed exactly, while a scalar quality model remains available as an independent check. It is therefore different from treating a Gaussian or scalar reward as the deployment objective.

There are two practical consequences. First, the five categories are the annotation protocol itself, so the upper endpoint and the gaps between ratings are not chosen after inspecting policy outcomes. Second, repeated labels make the conditional scale interpretable as disagreement in that protocol. We use this signal to diagnose whether a policy is relying on unstable evaluator regions, but never add it to the optimization target. The separation matters: using variance as a bonus would reward precisely the ambiguity that the calibration correction is intended to protect against.

The resulting interface is deliberately asymmetric. The mean is retained as an ordinary-quality floor, the variance as a diagnostic, and the categorical vector only where the finite-budget maximum needs it. This prevents the task from silently becoming uncertainty seeking. It also makes the artifact auditable: the predicted rating probabilities, their implied disagreement, and the independent quality score can be inspected separately instead of being collapsed into one scalar.

2.2 Finite-budget objective

For candidate j, write $F_{\omega,j}(t) = \sum_{r=0}^{t} p_{\omega,r}(x, Y_j)$. The nominal expected maximum has the exact finite-support form

$$h_N^{\mathrm{nom}}(x, Y_{1:N}) = \sum_{t=1}^{4} \left[1 - \prod_{j=1}^{N} F_{\omega,j}(t-1) \right]. \tag{1}$$

Thus rating noise is integrated out before policy optimization and no Gaussian or sampled-max approximation is introduced. The probability of at least one rating-four response is a secondary diagnostic; the complete ordinal maximum is the primary estimand (Appendix B).

The threshold form also clarifies what the policy is learning. A response is valuable when it reduces the probability that every other candidate remains below a threshold, not merely when its mean score is large. This gives a candidate credit that is sensitive to complementary upper-tail mass: two responses with the same mean can have different contributions if one changes the probability of crossing rating 3 or 4. A scalar max@32 baseline cannot represent this distinction because it ranks candidates only by their collapsed means.

This is where finite-budget optimization differs from ordinary reward maximization. The group does not need every response to be good; it needs at least one response to clear each relevant threshold. Improving a candidate that is redundant with the other (N-1) responses should therefore have little marginal value, while a candidate that supplies missing upper-tail mass should receive credit even if it is not the sampled winner. The closed form exposes this complementarity before any rollout rating is sampled.

This threshold view gives a precise comparison with scalar max training. Two responses can have the same μ_ω while placing their probability mass at different ordinal thresholds; the response with mass at a threshold that the other candidates do not cover has the larger q_i^{rob}. The distinction changes which response receives a policy gradient and remains visible when the group has no unique sampled winner. The ordinal vector is therefore used only where it carries decision-relevant information; PPO still consumes a single response-level advantage after this calculation.

2.3 Calibration uncertainty

Because the maximum magnifies upper-tail errors, we estimate a one-sided radius from independent prompt clusters before training. For threshold $t \in \{0, 1, 2, 3\}$, let $D_{g,t}$ be the within-cluster mean of $\mathbf{1}\{R \le t\} - F_\omega(t)$, and let $\widehat{d}_t$ be its mean across clusters. The frozen radius is

$$\varepsilon_{\mathrm{cal}} = \min\left\{ 1, \max_{t=0,\ldots,3} (\widehat{d}_t)_+ + \sqrt{\frac{\log(4/\alpha_{\mathrm{cal}})}{2G_{\mathrm{cal}}}} \right\}. \tag{2}$$

For $\mathcal{U}_\varepsilon(p) = \{q \in \Delta^4 : F_q(t) \le \min(1, F_p(t) + \varepsilon),\ t = 0, \ldots, 3\}$, let $\overline{F}_j(t) = \min(1, F_{\omega,j}(t) + \varepsilon_{\text{cal}})$, with $\overline{F}_j(-1) = 0$, $\overline{F}_j(4) = 1$, and induced mass $\overline{p}_{j,r} = \overline{F}_j(r) - \overline{F}_j(r-1)$. The robust objective is

$$\begin{aligned} h_N^{\text{rob}}(x, Y_{1:N}) &= \inf_{q_j \in \mathcal{U}_{\varepsilon_{\text{cal}}}(p_{\omega,j})} \mathbb{E}_q[\max_j R_j] = \sum_{t=1}^{4} \left[1 - \prod_{j=1}^{N} \overline{F}_j(t-1) \right], \\ J_N^{\text{rob}}(\theta) &= \mathbb{E}_{x, Y_{1:N}}[h_N^{\text{rob}}(x, Y_{1:N})]. \end{aligned} \tag{3}$$

The one-sided envelope is the key robustness choice: it adds adverse low-rating mass, but never converts variance into reward. Its radius is an explicit conservatism knob rather than a hidden tuning preference.

The direction of the envelope follows the deployment objective. For a maximum, increasing a candidate CDF decreases the chance of crossing every threshold; the upper-CDF envelope is therefore pessimistic in exactly the harmful direction. A symmetric interval would spend robustness budget on the benign direction as well and would no longer have the same closed-form worst case. Because the radius is fixed from a disjoint calibration split, this protection cannot be tuned toward a favorable policy after training.

The envelope is a robustness device, not a claim that calibration error is uniform across prompts. Its operational role is to make a specific failure mode explicit: if a response appears strong only because its predicted lower-tail probabilities are optimistic, the envelope removes that free advantage before the policy receives credit. Where the evaluator is well calibrated, the correction is small and the objective approaches the nominal finite-budget maximum. Thus ε_{cal} is a sensitivity knob, not an unexplained regularization coefficient.

2.4 Constrained policy objective

The maximum can improve a few candidates while degrading typical responses or over-optimizing the training evaluator. We therefore optimize J_N^{rob} subject to $M(\theta) = \mathbb{E}[\mu_\omega(x, Y)] \ge m_0$, $Q(\theta) = \mathbb{E}[Q_\eta(x, Y)] \ge q_0$, and $\mathbb{E}_x[\text{KL}(\pi_\theta \| \pi_{\text{ref}})] \le \delta$:

$$\max_\theta J_N^{\text{rob}}(\theta) \quad \text{s.t.} \quad M(\theta) \ge m_0, \quad Q(\theta) \ge q_0, \quad \mathbb{E}_x[\text{KL}(\pi_\theta \| \pi_{\text{ref}})] \le \delta. \tag{4}$$

The floors are fixed from the reference policy before training, while δ_{hard} is an empirical rollback threshold. The independent Quality RM and KL controller are safeguards against evaluator-specific overoptimization, not proofs that reward hacking is impossible.

Each constraint addresses a different failure mode. The nominal-mean floor prevents a tail optimizer from sacrificing the typical response; the independent Quality RM tests whether an improvement transfers beyond the training evaluator; and the KL budget limits how much behavior can change while these scores are being optimized. We consequently report the maximum and drift as separate axes instead of combining them into an arbitrary weighted sum.

The constraints are intentionally not folded into the robust maximum itself. Doing so would obscure whether an update improved the deployment target or merely paid for it with a different quality or drift trade-off. In the implementation, the floors enter through dual credits, the KL term enters the PPO surrogate, and rollback is a binary acceptance rule. The resulting diagnostics remain distinct: objective value, constraint residuals, measured held-out drift, and rejected-step frequency.

Relation to existing objectives. StableMax-PPO builds on PPO and prior inference-aware, pass@k, max@k, and Best-of-N policy-gradient objectives (Chow et al., 2025; Tang et al., 2025; Walder & Karkhanis, 2025; Bagirov et al., 2025; Li et al., 2026; Poiani et al., 2026); we do not claim the first finite-budget maximum. The new object is their combination with a bounded ordinal evaluator, exact robust order statistics, an exact marginal credit, and simultaneous mean, independent-quality, and KL safeguards.

3 METHOD

The method separates the deployment target from its stabilizers. The target is the robust ordinal maximum; the stabilizers control typical quality and policy motion. This separation makes the max–stability trade-off visible instead of hiding it in a scalar reward weight.

3.1 EXACT ROBUST MAXIMUM AND MARGINAL CREDIT

Theorem 1 (Exact finite-budget robust maximum)**.** *Under (A1), the envelope distribution $\overline{p}_j$ simultaneously attains the infimum in eq. (3) for every candidate. Hence the four-term expression is the exact worst-case expected maximum over the full declared product ambiguity set, not a Monte Carlo or Gaussian approximation.*

The theorem is useful computationally: it replaces an optimization over rating tables by four threshold products, so the robust target is evaluated without sampling latent ratings or enumerating 5^N outcomes. For a candidate i, let $\overline{R}_i \sim \overline{p}_i$ and $\overline{M}_{-i} = \max_{j \neq i} \overline{R}_j$. Its leave-one-out marginal credit is

$$q_i^{\mathrm{rob}} := \mathbb{E}[(\overline{R}_i - \overline{M}_{-i})_+ \mid x, Y_{1:N}] = \sum_{t=1}^{4} [1 - \overline{F}_i(t-1)] \prod_{j \neq i} \overline{F}_j(t-1). \tag{5}$$

This is the paper's central credit construction: it integrates out rating noise but preserves the response-level likelihood ratio, so the policy receives credit only for thresholds at which a candidate can improve the group maximum.

This credit is deliberately leave-one-out rather than winner-take-all. A winner indicator would give zero signal to a response that raises the chance of crossing a threshold without being the sampled winner, and would require an additional rating draw for every rollout. The product over $j \neq i$ measures how much room the other candidates leave at each threshold; the credit is thus both group-aware and response-specific. Prefix and suffix products evaluate all N credits in $O(N|\mathcal{R}|)$ time per prompt, which is the computational reason the exact construction is usable inside PPO.

The scaling statement matters for the engineering use case. A naive implementation would reevaluate a group maximum after perturbing each candidate, or enumerate rating tuples. Threshold-wise prefix and suffix products instead share intermediate values across all candidates. Exactness is therefore not only an analytical convenience: the deployment statistic and its training credit are computed by the same small tensor kernel.

Theorem 2 (Exact marginal policy credit)**.** *Under the regularity assumptions, q_i^{rob} satisfies*

$$\nabla_\theta J_N^{\mathrm{rob}}(\theta) = \mathbb{E}\left[\sum_{i=1}^{N} q_i^{\mathrm{rob}} \nabla_\theta \log \pi_\theta(Y_i \mid x)\right]. \tag{6}$$

Consequently, the response-level estimator is unbiased before PPO clipping.

The identity is more than a restatement of the maximum: it is the bridge from the robust deployment objective to a trainable update. It also avoids the high-variance choice of drawing a rating for every response and then assigning the sampled winner's credit.

The identity also explains why the proposed credit is response-level rather than token-level. The group statistic determines which response should move the policy, while the autoregressive likelihood ratio distributes that credit over its generated tokens. The method can therefore add a group-aware target without a new optimizer or a second rating-sampling loop. It remains compatible with PPO's clipping and minibatch machinery, while the exactness claim stays attached to the pre-clipping objective where it can be proved.

The identity is stated before clipping because it is a property of the target functional, not of a particular optimizer. PPO clipping may introduce the usual controlled surrogate bias, but it does not change which response-level quantity is being estimated. This separation lets the experiments attribute changes in the max–stability frontier to the proposed credit and controller, rather than to a hidden change in the deployment metric.

Implementation consequence. The estimator is exact with respect to the frozen evaluator and the declared product model, but it does not require sampling a latent rating. In a rollout, the same cached categorical probabilities feed the robust objective, every leave-one-out credit, and the diagnostic variance. This shared computation is important for engineering reproducibility: a mismatch between the metric used for evaluation and the credit used for training cannot be attributed to separate sampling noise or to an unreported reward transformation.

3.2 Constrained PPO update

For dual variables $\lambda_m, \lambda_q \geq 0$, the response-level credit used by PPO is

$$\widetilde{q}_i = q_i^{\text{rob}} + \frac{\lambda_m}{N}\mu_\omega(x, Y_i) + \frac{\lambda_q}{N}Q_\eta(x, Y_i), \qquad A_i = N[\widetilde{q}_i - b_\psi(x)]. \tag{7}$$

The factors $1/N$ match the one-response constraints to the group objective; the prompt-only baseline preserves the score-function identity. We use the standard clipped PPO surrogate and full-categorical token KL. Writing $\rho_{i,h} = \pi_\theta(a_{i,h} \mid s_{i,h})/\pi_{\text{old}}(a_{i,h} \mid s_{i,h})$, the update maximizes the standard per-token expression $\min\{\rho_{i,h}A_i, \text{clip}(\rho_{i,h}, 1 - \epsilon_{\text{PPO}}, 1 + \epsilon_{\text{PPO}})A_i\}$ averaged over candidates, minus $\beta_k\widehat{\text{KL}}$, and divides by the lagged RMS scale. The full implementation, including projected dual updates, is specified in Appendix C.

Before a tentative step, StableMax-PPO snapshots all policy parameters and the complete optimizer state. If the post-update KL on the same rollout states exceeds δ_{hard}, both snapshots are restored; baseline, dual, RMS, and controller states advance only after acceptance. Thus rollback is an exact batch-level invariant, while the population KL constraint remains an empirical design target.

The controller is intentionally conservative about state. A rejected update must not train the prompt baseline, change either dual variable, or alter the normalization statistics; otherwise the next accepted step would depend on an update that the algorithm itself declared unsafe. Snapshotting the optimizer state as well as the parameters is necessary for Adam-like optimizers, whose moment buffers otherwise retain the rejected step.

This state discipline makes the rollback count interpretable. A rejection is not a hidden partial update: the policy, optimizer moments, baseline, dual variables, and normalization scale all remain at the pre-step state. Accepted and rejected steps can therefore be compared across baselines under the same controller. The gate is still only a batch diagnostic, which is why we report both held-out KL and rollback frequency rather than presenting rollback as a population trust-region guarantee.

3.3 StableMax-PPO

Algorithm 1: StableMax-PPO update

Input: $\mathcal{D}, \pi_\theta, \pi_{\text{ref}}, p_\omega, Q_\eta, \varepsilon_{\text{cal}}, m_0, q_0, N, \delta_{\text{hard}}$
Output: Updated policy parameters θ.

```
Initialize b_ψ, λ_m, λ_q, β_KL and lagged scale s > 0;
for k = 1, ..., T_pol do
    Roll out N responses per prompt under π_old;
    Compute q̃_i and A_i from the closed-form credits;
    Snapshot (θ, Opt) and perform one clipped-PPO update;
    if measured post-update KL̂ > δ_hard then
        Restore (θ, Opt); leave controller state unchanged;
    else
        Commit and update b_ψ, λ_m, λ_q, β_KL, s;
```

Algorithm 1 is the only algorithm in the paper; the implementation constants and diagnostics are in the supplementary protocol.

4 Theoretical Analysis

The previous section gives the exact target and its policy-gradient bridge. Here we state the guarantees that explain the role of the calibration radius and the update controller. Proofs are in Appendix A.

4.1 Calibration coverage

Theorem 3 (Calibration-robust lower bound). *Let F_j^* be the true conditional CDF. If $F_j^*(t) \leq \overline{F}_j(t)$ for every candidate and $t \in \{0, 1, 2, 3\}$, then the true conditional maximum satisfies $\mathbb{E}_*[\max_j R_j \mid x, Y_{1:N}] \geq h_N^{\mathrm{rob}}(x, Y_{1:N})$. Without exact coverage, the gap is at most $4N\xi(x, Y_{1:N})$, where $\xi = \max_{j,t}[F_j^*(t) - \overline{F}_j(t)]_+$.*

This theorem makes the claim deliberately conditional: calibration on prompt clusters is not a pointwise guarantee for every generated response. The envelope is valuable because it converts that uncertainty into a measurable lower-tail protection term rather than an informal confidence statement. In particular, the theorem does not say that the frozen Moment RM is correct on every new response. It says what follows if the declared envelope covers the relevant conditional CDFs, and quantifies the penalty when it misses. This is the distinction between a calibration-robust objective and an unsupported claim of reward-model reliability.

4.2 Dependence on the calibration radius

Proposition 1 (Calibration-radius monotonicity). *For $0 \leq \varepsilon_1 \leq \varepsilon_2 \leq 1$, $0 \leq J_N^{\mathrm{rob}}(\theta; \varepsilon_1) - J_N^{\mathrm{rob}}(\theta; \varepsilon_2) \leq 4N(\varepsilon_2 - \varepsilon_1)$, and at $\varepsilon = 0$, $J_N^{\mathrm{rob}}(\theta; 0) = J_N^{\mathrm{nom}}(\theta)$.*

The radius is therefore an interpretable robustness knob: increasing it can only make the target more conservative, with a finite sensitivity bound.

The bound also clarifies how the knob should be used. It is a statement about the target before policy optimization: increasing the radius cannot create a spurious gain in the reported robust maximum, although it can change which responses receive credit. This is different from tuning a KL coefficient, which changes the optimization path while leaving the deployment functional fixed. Reporting both quantities lets the experiment distinguish conservatism in the objective from conservatism in the update.

4.3 Rollback invariant

Proposition 2 (Rollback correctness). *If a tentative update is rejected, restoring the parameter and optimizer snapshots exactly returns the measured rollout-state policy and optimizer to their pre-update values. This is a batch-level invariant, not a population KL guarantee.*

The three results separate the mathematical and engineering roles of the method: calibration protects the target, the marginal credit makes it optimizable, and rollback controls observed policy motion.

Together, these statements provide a compositional account of the method's value. The finite-support result is exact under the declared independence model; the coverage result states what calibration error can and cannot imply; and rollback gives an implementation invariant for the observed batch. None of them is a blanket claim of human alignment or a population trust region, which is why the empirical section treats the safeguards as measurable operating-point controls.

The guarantees are intentionally modular. The finite-support and gradient results are properties of the evaluator interface and the deployment functional, whereas rollback is a property of the optimizer state machine. Improving one component therefore cannot silently repair a failure in another: a stable update does not correct a calibration miss, and a calibrated objective does not prevent a large policy step. This separation is what makes the calibration residual, rollback count, and independent-evaluator diagnostics useful as distinct engineering signals.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

The policy is Qwen2.5-1.5B-Instruct. The frozen ordinal Moment RM and independent Quality RM are Qwen2.5-14B models trained with HelpSteer2 repeated helpfulness ratings and preference pairs (Wang et al., 2024b). We use three confirmatory seeds $\{314, 2718, 1618\}$, $N = 32$, and 512 held-out prompts per seed. Primary baselines are Mean-PPO + Q, GRPO + Q, and Scalar Max@32-PPO + Q; full settings, ablations, and gates are in Appendix B. We report robust maximum, held-out KL, and rollback frequency because they measure the two sides of the proposed operating point.

5.2 MAIN RESULTS

StableMax-PPO reaches a robust expected maximum of 3.458 and held-out KL of 0.034, compared with 3.424 and 0.070 for Mean-PPO. Under the shared rollback controller it rejects 31.0% of tentative updates, versus 58.6% for Mean-PPO and 82.3% for GRPO. GRPO and Scalar max-PPO attain higher robust maxima, so the evidence supports a lower-drift operating point rather than universal dominance.

Table 1: Primary aggregate results (three seeds, 512 prompts per seed).

Method	Robust max	95% CI	Held-out KL	Rollbacks/900
StableMax-PPO	3.458	[3.411, 3.506]	0.034	279 (31.0%)
Mean-PPO	3.424	[3.354, 3.517]	0.070	527 (58.6%)
GRPO	3.555	[3.530, 3.578]	0.094	741 (82.3%)
Scalar max-PPO	3.517	[3.476, 3.558]	0.054	285 (31.7%)
Best-of-32 (base)	3.086	[3.045, 3.125]	–	–

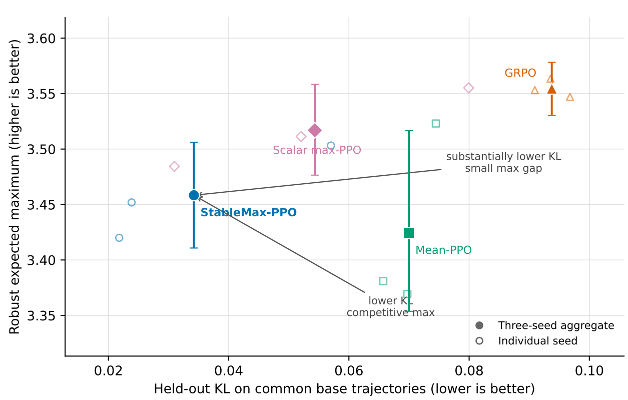

Figure 1: Robust maximum versus held-out KL. Filled markers are three-seed aggregates; open markers are individual seeds and bars are 95% intervals.

Figure 1 makes the trade-off explicit: StableMax-PPO has lower held-out KL than Mean-PPO and GRPO while retaining a competitive robust maximum, whereas max-oriented comparators remain higher on that endpoint. Rollback frequency is an optimization diagnostic, not a population guarantee.

The aggregate numbers quantify the operating point rather than a universal ranking. Relative to Mean-PPO, StableMax-PPO increases robust maximum by 0.034 while reducing held-out KL from 0.070 to 0.034 (a 51% relative reduction). Relative to Scalar max-PPO, it gives up 0.059 robust-max points while using 0.020 less KL. These are the two axes that the method exposes; collapsing them into one weighted score would conceal the engineering trade-off the experiment is designed to measure.

5.3 Ablation and stability analysis

Removing the mean floor increases robust max to 3.508 but raises held-out KL to 0.048; removing the independent quality floor yields 3.467 at KL 0.048. Removing calibration robustification yields 3.421 at KL 0.028. These ablations move along the same frontier and support interpreting the constraints as operating-point controls, not uniformly beneficial reward terms. Per-seed values and the remaining diagnostics are in the appendix.

Table 2 makes the component interpretation explicit. Removing the mean floor buys maximum performance at a measurable drift cost, while removing calibration changes the target itself and therefore cannot be interpreted as a pure optimizer ablation. The independent Quality floor has a smaller endpoint effect but still changes the accepted-update trajectory. These comparisons are why we present StableMax-PPO as a controlled operating point: each safeguard has a distinct failure mode and none should be described as a universally improving reward term.

Table 2: Component ablations (three-seed aggregate). Values are unchanged from the complete supplementary table.

Variant	Robust max	Held-out KL	Rollbacks/900
StableMax-PPO	3.458	0.034	279 (31.0%)
No mean floor	3.508	0.048	230 (25.6%)
No quality floor	3.467	0.048	329 (36.6%)
No calibration correction	3.421	0.028	121 (13.4%)

6 Limitations

The endpoint is induced by frozen evaluators and no fresh human labels are collected for generated outputs. Model proxies may disagree with humans, the 1.5B/14B capacity gap may induce evaluator shift, and three seeds do not establish scale or domain generalization. The exact product form also assumes conditionally independent rating outcomes across sampled responses and does not model correlated annotation or evaluator errors. The artifact reports these limitations and all unfavorable gates rather than redefining the protocol.

Reproducibility statement. The supplementary artifact contains the source, official ICLR style files, protocol, result tables, verification JSON, build commands, and the anonymous repository at `https://github.com/LoveGuys-cmd/rlhf-artifact`. The main paper specifies the objective, algorithm, seeds, metrics, and aggregate results; appendices give proofs and the complete protocol. Model weights, credentials, private data, and other non-redistributable files are excluded.

AI use statement. Generative AI tools assisted with paper organization, methodological exposition, language editing, LaTeX and figure formatting, and debugging of analysis code. The authors independently checked all mathematical claims and proofs, verified implementation and numerical values against frozen result files, and reviewed every AI-assisted change. No AI system generated experimental data or replaced author verification.

Ethics statement. The study uses model-generated text and existing preference data. It collects no new human-subject data or personal information. Model-based preference proxies are reported as proxies, not as human judgments of newly generated outputs; possible reward-model exploitation and evaluator shift are disclosed in Section 6 and Appendix B.

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A DETAILED THEORETICAL ANALYSIS

A.1 ORDINAL-PROBIT PARAMETERIZATION

The frozen Moment RM emits a latent location $m_\omega(x, y)$ and positive scale $s_\omega(x, y)$. With frozen cutpoints $c_1 < \cdots < c_4$, its five-category probabilities are $p_{\omega,0} = \Phi((c_1 - m_\omega)/s_\omega)$, $p_{\omega,r} = \Phi((c_{r+1} - m_\omega)/s_\omega) - \Phi((c_r - m_\omega)/s_\omega)$ for $r = 1, 2, 3$, and $p_{\omega,4} = 1 - \Phi((c_4 - m_\omega)/s_\omega)$. The moments used by the policy are $\mu_\omega = \sum_{r=0}^{4} r p_{\omega,r}$ and $\sigma_\omega^2 = \sum_{r=0}^{4} (r - \mu_\omega)^2 p_{\omega,r}$. Repeated-rating likelihood and disagreement supervise the conditional scale; therefore this variance is a model-based signal of annotation disagreement, not epistemic uncertainty or a stand-alone reward. Proper scoring and calibration are required because the complete categorical distribution enters the finite-budget objective (McCullagh, 1980; Kendall & Gal, 2017; Lou et al., 2024; Sun et al., 2025; Gneiting & Raftery, 2007; Guo et al., 2017).

This parameterization is included to make the evaluator interface reproducible, not to claim a new ordinal-link model. The methodological choice is what the policy does with the frozen output: the same five probabilities support the nominal mean, threshold products, robust envelope, and disagreement diagnostic. Keeping these uses separate is important because replacing the vector by its mean would make the finite-budget objective and its calibration check unidentifiable.

Proposition 3 (Finite-support threshold decomposition). *For conditionally independent ratings supported on $\mathcal{R} = \{0, 1, 2, 3, 4\}$, eq. (1) is exact, lies in $[0, 4]$, and is nondecreasing under first-order stochastic improvement of any candidate. Moreover, eq. (8) is its highest-threshold component.*

The proposition is the finite-support sanity check for the main construction. It also explains why the top-rating diagnostic in the artifact is not a substitute for the reported objective: it is only the $t = 4$ term, whereas the method trains against all four threshold events. This distinction prevents a policy from appearing successful by concentrating probability only at the highest category while degrading the intermediate ordinal structure.

Proof. For any integer-valued $Z \in \{0, \ldots, 4\}$, the tail-sum identity gives $\mathbb{E}[Z] = \sum_{t=1}^{4} \Pr\{Z \geq t\}$. Taking $Z = \max_j R_j$ and using (A1) gives $\Pr\{\max_j R_j < t \mid x, Y_{1:N}\} = \prod_{j=1}^{N} F_{\omega,j}(t-1)$.

Substitution proves eq. (1). Each term lies in $[0, 1]$, so the sum lies in $[0, 4]$. First-order stochastic improvement decreases a candidate CDF and therefore cannot decrease any summand. At $t = 4$, $F_{\omega,j}(3) = 1 - p_{\omega,4}(x, Y_j)$, giving eq. (8). □

Proposition 4 (Prompt-cluster calibration statement). *Assume the validation prompt clusters are independent and that the RM is fixed before evaluating them. Then the radius in eq. (2) satisfies, with probability at least $1 - \alpha_{\text{cal}}$, $\mathbb{E}[D_t] \leq \varepsilon_{\text{cal}}$ simultaneously for $t = 0, 1, 2, 3$.*

Proof. Conditional on the frozen responses and predictions, changing all binary indicators in a prompt cluster from zero to one changes $D_{g,t}$ by at most one. Thus each cluster residual lies in an interval of width one. Hoeffding's inequality gives $\Pr(\mathbb{E}[D_t] - \widehat{d}_t > \sqrt{\log(4/\alpha_{\text{cal}})/(2G_{\text{cal}})}) \leq \alpha_{\text{cal}}/4$.

A union bound over four thresholds proves the simultaneous event. On that event, every right-hand side is bounded by the unclipped expression in eq. (2). If clipping is active, $\varepsilon_{\text{cal}} = 1$, while $\mathbb{E}[D_t] \leq 1$, so the claim still holds. □

Scope of the calibration proposition. Proposition 4 is a population-average statement over the validation prompt distribution. It is not a distribution-free, pointwise guarantee that every conditional CDF lies in the ambiguity set. The pointwise premise required by the next theorem is therefore stated explicitly rather than inferred from a small number of repeated ratings.

This scope distinction is operationally relevant. The calibration split controls how the radius is chosen before training, while the theorem in the main text describes what follows if the resulting envelope covers a generated candidate. Conflating these statements would turn an average calibration diagnostic into an unjustified pointwise claim; the artifact keeps the calibration estimate and the conditional robustness guarantee as separate records.

Proof of Theorem 1. The envelope $\overline{F}_j$ is nondecreasing, has boundary values zero and one, and therefore defines a valid categorical distribution. For every feasible $q_j \in \mathcal{U}$, $F_{q_j}(t) \le \overline{F}_j(t)$ for $t = 0, 1, 2, 3$.

Hence q_j first-order stochastically dominates $\overline{p}_j$, or equivalently $\overline{p}_j$ is stochastically smallest. The maximum is coordinatewise nondecreasing in its ratings, so replacing every feasible candidate by $\overline{p}_j$ cannot increase its expectation. The envelope itself is feasible and thus attains the infimum. Applying proposition 3 gives the stated closed form.

Proof of Theorem 3. Let F_j^* be the true conditional CDF. Under the pointwise premise $F_j^*(t) \le \overline{F}_j(t)$ for every candidate and threshold, first-order stochastic dominance gives the lower-bound inequality in Theorem 3. For the misspecified case, define $\xi = \max_{j,t}[F_j^*(t) - \overline{F}_j(t)]_+$. At each threshold, the true and robust tail-sum terms can differ only through the corresponding CDF products. For $a_j, b_j \in [0,1]$, $\prod_j a_j - \prod_j b_j \le \sum_j (a_j - b_j)_+$.

Apply this inequality with $a_j = F_j^*(t-1)$ and $b_j = \overline{F}_j(t-1)$ at each of four thresholds. The discrepancy is at most $N\xi$ per threshold and therefore at most $4N\xi$ after summation. Taking outer expectations preserves the resulting inequality $\mathbb{E}_*[\max_j R_j] \ge h_N^{\mathrm{rob}} - 4N\xi$.

Proof of Proposition 1. Fix a candidate group and threshold t, and write $a_j(\varepsilon) = \min\{1, F_{\omega,j}(t-1) + \varepsilon\}$. For $\varepsilon_1 \le \varepsilon_2$, the clipping map gives $0 \le a_j(\varepsilon_2) - a_j(\varepsilon_1) \le \varepsilon_2 - \varepsilon_1$. The telescoping product identity and $a_j(\varepsilon) \in [0,1]$ then give $0 \le \prod_j a_j(\varepsilon_2) - \prod_j a_j(\varepsilon_1) \le \sum_j (a_j(\varepsilon_2) - a_j(\varepsilon_1)) \le N(\varepsilon_2 - \varepsilon_1)$. Substituting this bound into the four-term tail-sum expression for h_N^{rob}, and then taking the expectation over prompts and responses, proves Proposition 1. At $\varepsilon = 0$, the envelope equals the nominal CDF, so the exact infimum in Theorem 1 equals $J_N^{\mathrm{nom}}(\theta)$.

Proof of Proposition 2. Let $(\theta_0, \mathsf{Opt}_0)$ be the snapshot taken before a tentative update. If the measured rollout-state KL exceeds δ_{hard}, the algorithm assigns $(\theta, \mathsf{Opt}) \leftarrow (\theta_0, \mathsf{Opt}_0)$, so both objects equal their pre-update values exactly (up to the semantics of the implementation's state-copy operation). Because controller-state updates occur only in the accepted branch, a rejected transition also leaves the baseline, dual variables, KL coefficient, and RMS scale unchanged. The statement is conditional on the measured batch and therefore makes no population-level KL claim.

Proof of Theorem 2. For nonnegative integer ratings, $(\overline{R}_i - \overline{M}_{-i})_+ = \sum_{t=1}^4 \mathbf{1}\{\overline{R}_i \ge t, \overline{M}_{-i} < t\}$. Taking conditional expectations and using (A1) gives $\Pr\{\overline{R}_i \ge t, \overline{M}_{-i} < t \mid x, Y_{1:N}\} = [1 - \overline{F}_i(t-1)] \prod_{j \ne i} \overline{F}_j(t-1)$,

which proves eq. (5). For the gradient, fix i and condition on (x, Y_{-i}). The group value decomposes as $h_N^{\mathrm{rob}} = \mathbb{E}[\overline{M}_{-i} \mid x, Y_{-i}] + q_i^{\mathrm{rob}}$.

The first term is independent of Y_i, and the score-function identity implies that its product with $\nabla_\theta \log \pi_\theta(Y_i \mid x)$ has conditional expectation zero. Applying the likelihood-ratio derivative to all N iid responses and summing the remaining terms proves eq. (6).

Proposition 5 ($N = 1$ reduction and complexity)**.** *For $N = 1$, $h_1^{\mathrm{rob}} = q_1^{\mathrm{rob}} = \sum_{r=0}^4 r\overline{p}_{1,r}$.*

Thus the upper-tail reward component reduces exactly to robust mean reward. For general N, all group values and marginal credits can be computed in $O(N|\mathcal{R}|)$ time using threshold-wise prefix and suffix products.

This reduction is a useful implementation check: at $N = 1$, the method must recover the robust mean and cannot manufacture a group advantage. For $N > 1$, the additional terms are exactly the response's incremental chance of opening a new threshold above the other candidates. The test suite evaluates both regimes, so a change in the group objective cannot be explained by a special-case implementation.

Proof. With one candidate, the empty opponent maximum is zero, so $(\overline{R}_1 - 0)_+ = \overline{R}_1$. The first equality follows. For complexity, each threshold requires one forward and one backward product scan; there are $|\mathcal{R}| - 1 = 4$ nontrivial thresholds. $\square$

B FULL EXPERIMENTAL PROTOCOL

B.1 EXPERIMENTAL QUESTIONS

The experiments are designed to answer four predeclared questions:

1. Does StableMax-PPO improve the held-out robust $\mathbb{E}[\max_{j \le 32} R_j]$ objective relative to mean optimization and a direct scalar max@32 policy gradient?
2. Are gains preserved under the nominal exact ordinal maximum and the top-rating probability $\Pr\{\exists j : R_j = 4\}$?
3. Do nominal expected rating, independent Quality RM score, and reference-policy KL remain within their predeclared safeguards?
4. Do independent generated-output proxies and existing human-labeled preference diagnostics provide evidence against target-RM exploitation?

No result is used to remove a comparator, redefine a gate, or select a more favorable protocol.

B.2 GATE OUTCOMES AND SCIENTIFIC INTERPRETATION

The artifact verifier passed for seeds 314, 2718, and 1618. Passed gates are the ArmoRM generated-output proxy, held-out KL superiority versus Mean-PPO and GRPO, the nominal-mean floor, and the independent Quality-RM floor. Failed gates are the Qwen generated-output proxy, robust non-inferiority versus Mean-PPO and GRPO, and the primary comparison versus Scalar Max-PPO. Accordingly, the preregistered all-gates field is `scientific_success=false`. This records a max–stability trade-off; it is not evidence of direct human-preference improvement or a proof that reward-model exploitation is absent.

The gate vector is intentionally read as a decision record rather than a single leaderboard score. A method can pass the KL and quality safeguards while failing a max non-inferiority comparison, or improve the target reward while failing an independent proxy. Reporting the vector preserves these distinctions and prevents a favorable endpoint from hiding a failure mode that the method was designed to measure.

B.3 MODELS, DATA, AND SPLIT DISCIPLINE

The policy is Qwen2.5-1.5B-Instruct (Yang et al., 2024). The frozen Moment RM is a Qwen2.5-14B ordinal model trained from HelpSteer2 repeated helpfulness ratings and preference pairs (Wang et al., 2024b). It exports $p_{\omega,0:4}$ directly; exposing the category probabilities does not retrain or alter its accepted checkpoint. An independently trained Qwen2.5-14B scalar Quality RM supplies Q_η and has no variance head. Its frozen 4,096-pair confirmation accuracy is 94.60% with a Wilson lower 95% bound of 93.87%.

All policy, reward-model, evaluator, and lockbox splits are content-hashed. Before partitioning RewardBench, we remove prompt and response overlap with the policy and Quality-RM data (Lambert et al., 2024). The remaining calibration, policy-diagnostic, and reserve partitions are also mutually prompt- and response-disjoint. They contain 1,024, 1,024, and 427 pairs, respectively. The sealed Skywork diagnostic contains 4,096 pairs and is excluded from Quality-RM training and selection (Liu et al., 2024).

The Moment RM and policy prompts remain drawn from the same broad HelpSteer2 domain. Disjoint rows prevent direct leakage but do not establish out-of-domain generalization; this is treated as a limitation rather than hidden by the split terminology.

B.4 Frozen training protocol

The primary response budget is $N = 32$. Each trainable confirmatory method uses three seeds $\{314, 2718, 1618\}$, 300 policy steps, two prompts per step, 32 responses per prompt, learning rate 2×10^{-5}, one PPO epoch, clip 0.2, LoRA rank 8 with scale 32, sampling temperature 1, $\mathrm{top}_p = 1$, maximum prompt length 384, and maximum response length 64. Seed 42 is used only for development and real-GPU smoke tests and is excluded from all confirmatory intervals, gates, and aggregate claims. The target KL is 0.02 and the hard rollback level is 0.04. All methods share the same prompt stream, generation budget, optimizer class, policy architecture, and checkpoint-selection rule.

The non-inferiority floors are frozen before any trainable policy outcome is observed. The reference sample contains 2,048 base-policy responses from 512 independent policy-training prompts, with four responses per prompt. For $Z \in \{\mu_\omega, Q_\eta\}$, let $\bar{Z}_g$ be the four-response prompt mean and let L_Z be the fifth percentile of 10,000 bootstrap means obtained by resampling prompts. The frozen floor is $z_0 = L_Z - 0.1 s_Z$, where s_Z is the response-level standard deviation in the same reference sample. The bootstrap unit, one-sided level, 10,000 draws, sample counts, generation seed, bootstrap seed, and practical margin are all predeclared. A separate prompt-disjoint sample of 512 responses from 128 prompts reports standardized reference-policy shift. It is diagnostic only and cannot accept or reject training.

B.5 Comparators and ablations

The ten trainable methods are:

1. **StableMax-PPO**: the complete method in Algorithm 1.
2. **Mean-PPO + Q**: PPO on μ_ω with the same mean, Quality, and KL protocol.
3. **GRPO + Q**: group-relative standardized μ_ω credit with matched safeguards (Shao et al., 2024; Guo et al., 2025).
4. **Scalar Max@32-PPO + Q**: exact leave-one-out scalar max credit $[\mu_i - \max_{j \neq i} \mu_j]_+$ with matched safeguards (Tang et al., 2025; Walder & Karkhanis, 2025; Bagirov et al., 2025).
5. **Entropic-PPO + Q**: PPO on $\mathrm{CE}_\tau = -\tau \log \sum_r p_{\omega,r} e^{-r/\tau}$, using $\tau = 1$.
6. **Nominal Ordinal EV-PPO + Q**: exact J_{32}^{nom} without the calibration ambiguity set.
7. **StableMax-PPO without mean floor**: ablates nominal-mean non-inferiority.
8. **StableMax-PPO without Quality RM**: ablates the independent quality safeguard and is excluded from constrained-feasibility claims.
9. **Gaussian EV-PPO + Q**: the previous heteroscedastic Gaussian expected-maximum objective, retained as a model-assumption ablation.
10. **Top-4 PPO + Q**: exact optimization of $\Pr\{\exists j : R_j = 4\}$, isolating the highest threshold from the complete ordinal maximum.

An eleventh, evaluation-only **Best-of-32** row samples 32 responses from the untrained base policy and ranks them by μ_ω. It is an inference-time reference, not a compute-matched training baseline; its generation and scoring cost is reported separately.

The comparator set is organized to separate three changes that are otherwise easy to conflate: the deployment functional (mean, scalar maximum, top-rating, or full ordinal maximum), the calibration correction, and the stabilizers. The nominal ordinal and Gaussian variants hold the policy optimizer fixed while changing the evaluator assumption; the no-floor variants hold the target fixed while removing one safeguard. This is why the ablations support a causal interpretation of the observed frontier without claiming that any single component is universally beneficial.

B.6 Evaluation endpoints

Every method is evaluated on 512 frozen test prompts with newly generated groups of 32 responses. The primary paired prompt-level endpoint is h_{32}^{rob}, exactly matching the StableMax-PPO training objective. Secondary model-based endpoints are h_{32}^{nom}, U_{32}^{nom}, candidate-level μ_ω, ordinal variance,

Table 3: Complete aggregate table. Values are three-seed robust expected maximum, with 95% confidence intervals, held-out KL, and rollback counts.

Method	Robust max	95% CI	Held-out KL	Rollbacks/900
StableMax-PPO	3.458	[3.411, 3.506]	0.034	279 (31.0%)
Mean-PPO	3.424	[3.354, 3.517]	0.070	527 (58.6%)
GRPO	3.555	[3.530, 3.578]	0.094	741 (82.3%)
Scalar max-PPO	3.517	[3.476, 3.558]	0.054	285 (31.7%)
Entropic PPO	3.400	[3.353, 3.447]	0.037	302 (33.6%)
Nominal EV-PPO	3.421	[3.377, 3.467]	0.028	121 (13.4%)
StableMax-PPO (no mean floor)	3.508	[3.426, 3.574]	0.048	230 (25.6%)
StableMax-PPO (no quality floor)	3.467	[3.402, 3.530]	0.048	329 (36.6%)
Gaussian EV-PPO	3.558	[3.533, 3.581]	0.061	367 (40.8%)
Top-4 PPO	3.418	[3.353, 3.499]	0.042	277 (30.8%)
Best-of-32 (base)	3.086	[3.045, 3.125]	–	–

$p_{\omega,4}$, independent Quality score, reference KL, constraint residuals, accepted updates, and exact rollback counts.

Sensitivity at $N \in \{1, 2, 4, 8, 16, 32\}$ uses nested prefixes of the same cached 32-response groups. This preserves paired comparisons and avoids regenerating a favorable sample for each N. The group-level maximum uses all candidates and does not require a selected response. A deployment-style $\arg\max_i \mu_\omega(x, Y_i)$ response is reported only as a secondary selector diagnostic.

The nested-N evaluation is also a variance-control choice. Reusing the same cached response prefixes makes changes in the endpoint attributable to the budget rather than to a new generation sample. The held-out KL is computed on common base trajectories for the same reason, so a lower value reflects less measured policy motion under a paired protocol rather than an easier test set.

For completeness, the top-category diagnostic used in the cached evaluation is

$$U_N^{\mathrm{nom}}(\theta) := \mathbb{E}_{x,Y_{1:N}}\left[1 - \prod_{j=1}^{N}(1 - p_{\omega,4}(x, Y_j))\right]. \tag{8}$$

It is the frozen-model probability that at least one candidate receives rating four. It is not a Gaussian tail approximation and is not the primary estimand; the complete ordinal maximum remains the reported objective.

B.7 STATISTICAL ANALYSIS AND SUCCESS CRITERIA

All seeds are reported separately. Aggregate uncertainty is estimated by a hierarchical bootstrap that resamples seeds and prompts. Paired prompt-level contrasts additionally report one-sided sign-flip randomization tests and matched-pairs rank-biserial effect sizes. The primary superiority gates require the lower endpoint of a two-sided 95% hierarchical-bootstrap interval for the StableMax-PPO improvement in h_{32}^{rob} to exceed zero against Mean-PPO, GRPO + Q, and Scalar Max@32-PPO.

Safeguard success requires the corresponding interval lower endpoints for StableMax-PPO's candidate nominal mean and independent Quality mean to remain at or above m_0 and q_0. A point estimate alone cannot satisfy either floor. The complete scientific-success decision requires all predeclared superiority, floor, and generated-output proxy gates; improvement under the target RM alone is insufficient.

B.8 ANTI-EXPLOITATION AND HUMAN-GROUNDED DIAGNOSTICS

Newly generated outputs are compared with Mean-PPO outputs by two frozen evaluators: a Qwen2.5-14B judge and ArmoRM (Zheng et al., 2023; Wang et al., 2024a). Each evaluator must first pass its own frozen RewardBench calibration gate. Their pairwise judgments are model-based proxies, not human labels. Reward-hacking diagnostics compare target-RM improvements with proxy preference, response length, repetition, diversity, refusal behavior, and upper-tail strata. Target-RM improvement accompanied by proxy degradation is reported as evidence consistent with reward-model exploitation.

RewardBench and Skywork chosen–rejected lockboxes contain previously collected human labels. For each policy, length-normalized chosen–rejected policy log-probability accuracy and margin are reported as preference-retention diagnostics. They are not Best-of-32 endpoints and do not label newly sampled policy responses. In particular, a new response cannot be declared human preferred merely because it was generated for a prompt that also appears in a fixed preference benchmark.

No new human annotation is collected in this protocol. Therefore, even if all anti-exploitation gates pass, the permitted claim is *no evidence of exploitation under the measured model-based and existing-label diagnostics*. The experiment cannot establish absence of reward hacking or direct human preference for generated outputs. A blinded group-rating packet is emitted so that fresh human evaluation can be added later without regenerating or selecting outputs.

B.9 Reproducibility and reporting

Code, model adapters, data partitions, floor samples, calibration reports, and protocol files are content-hashed before training. A real-GPU smoke test must execute every declared objective, verify finite applicable metrics, verify constraint activation, and test exact parameter and optimizer rollback. Formal training and evaluation run only after these checks pass. Technical failures are versioned and corrected with regression tests. Unfavorable scientific outcomes are reported without changing seeds, thresholds, comparators, evaluators, or success criteria.

The 1.5B policy is evaluated by 14B reward and proxy models. This capacity gap may improve discrimination but may also create systematic evaluator shift. The three-seed design measures optimization variability; it does not replace replication at other model scales or fresh blinded labels.

C Implementation Details for StableMax-PPO

This appendix records implementation conventions for Algorithm 1; it is not a second algorithm. Model identifiers, seeds, and acceptance thresholds are specified in Appendix B.

For each prompt x_b, the implementation caches behavior-policy token log probabilities and response masks before evaluating any reward model. The five ordinal probabilities are converted to prefix CDFs. For $t = 0, \ldots, 3$, the robust envelope is $\bar{F}_{b,i}(t) = \min\{1, F_{\omega,b,i}(t) + \varepsilon_{\mathrm{cal}}\}$, with $\bar{F}_{b,i}(-1) = 0$ and $\bar{F}_{b,i}(4) = 1$. The robust probabilities are adjacent differences of this envelope. Prefix and suffix products compute all group values and leave-one-out credits in $O(BN|\mathcal{R}|)$ time without enumerating rating tuples.

The constrained credit used by the PPO surrogate is $\tilde{q}_{b,i} = q^{\mathrm{rob}}_{b,i} + \lambda_m \mu_\omega(x_b, Y_{b,i})/N + \lambda_q Q_\eta(x_b, Y_{b,i})/N$, with $A_{b,i} = N[\tilde{q}_{b,i} - b_\psi(x_b)]$. The prompt baseline is fitted only from accepted transitions. Dual variables use projected updates after acceptance; rejected steps leave all controller state unchanged. The lagged RMS scale is computed from accepted credits only, so a rejected outlier cannot change the next normalization factor.

Before the optimizer step, the implementation snapshots all trainable policy parameters and the complete optimizer state, including momentum and adaptive variance buffers. It evaluates full-categorical token KL on the same rollout states after the tentative step. If the hard threshold is exceeded, both snapshots are restored and no baseline, dual, KL-controller, or RMS state is advanced. This is an exact batch-level rollback invariant; it does not imply a population KL bound.

All floating-point reductions use deterministic tensor order where supported. The supplementary code contains tests for ordinal identities, credit reductions, finite metrics, protocol hashes, and rollback restoration.

D Artifact and verification records

The machine-readable supplementary files include the aggregate JSON, complete per-seed CSV tables, terminal verification records, and SHA-256 manifest.